

Contextual Sparse Embedded Topic Modeling for Mass2Motif Discovery

Abstract

Mass2Motifs are recurring fragment and neutral-loss patterns discovered by treating tandem mass spectra as short documents. Neural topic models can use one shared encoder to infer topic mixtures for every spectrum, but short spectra require mixtures that are simultaneously sparse and supported by a broad global topic inventory. We introduce Contextual Sparse ETM for this regime. The model starts from the published Embedded Topic Model (ETM): an explicit topic-word generator in a shared embedding space and a neural encoder that parameterizes a Gaussian posterior for each spectrum. We use fixed train-only spectral-word coordinates, balance fragment and neutral-loss probability mass, add contextual top-2 token-to-topic evidence to the posterior, and replace the dense document-topic softmax with 1.5-entmax. The adaptations add one learned scalar to a 19,278,000-parameter ETM.

The method was evaluated on a positive-ion-mode MSⁿLib reference-spectral benchmark partitioned so that identical compounds and molecular scaffolds do not cross splits. It used 1000 topics, 27222 training spectra and 3889 validation spectra. The model produced 803 motifs retained by Mass2Motif Annotation Guidance (MAG), 445 with an evaluable substructure overlap score (SOS), and 289 useful motifs with SOS at least 0.6. The corresponding Tomotopy latent Dirichlet allocation (LDA) counts were 607, 206 and 138. Held-out half-spectrum completion negative log-likelihood (NLL) was 9.5429 versus 9.6622, whereas Tomotopy had higher conditional mean SOS averaged over evaluable motifs (0.6761 versus 0.6472). Across three training seeds on the same frozen split, evaluable and useful motif counts remained within 439–453 and 274–289, median effective topics remained approximately 3.7, and 813–828 topics became the largest component for at least one validation spectrum. These results support a reproducible breadth-quality trade-off: the method discovers a substantially larger usable motif inventory while preserving sparse spectrum assignments. The held-out test partition was not accessed in this validation study.

1 Introduction

MS2LDA represents tandem mass spectra as documents, fragment and neutral-loss features as words, and recurrent co-occurrence patterns as latent topics called Mass2Motifs [1]. A molecule may contain several reusable substructures, so a spectrum should be represented by a mixture of motifs rather than assigned to a single cluster. This probabilistic representation supports both exploratory substructure discovery and downstream chemical annotation [2].

Mass spectra are an unusually demanding document regime. A spectrum contains a small number of physical peaks, each peak can generate paired fragment and neutral-loss words, and normalized intensities are encoded as pseudo-counts. At the same time, large reference libraries motivate topic inventories with hundreds or thousands of components. A useful model must therefore meet three requirements at once: predict withheld spectral evidence, assign each short spectrum to only a few motifs, and retain many distinct topics across the corpus. Likelihood alone is insufficient because diffuse document mixtures or a small effective topic inventory can coexist with good prediction.

The Embedded Topic Model (ETM) is the base model used in this study. It retains an explicit topic-word distribution, represents topics and words in a shared embedding space, and replaces iterative document-level inference with a neural variational encoder [7]. Spectral-word embeddings are also well motivated in mass spectrometry: Spec2Vec, a learned mass-spectral similarity method, demonstrated that fragment and neutral-loss co-occurrence embeddings capture chemically useful relationships [13]. However, an ordinary ETM posterior does not explicitly use the local co-occurrence pattern surrounding each word in a short spectrum, and softmax cannot produce exact zero topic probabilities.

Contextual Sparse ETM extends that published ETM in three explicit steps. First, the decoder assigns equal total probability to fragment and neutral-loss channels so their unequal vocabulary sizes cannot set their relative mass. Second, each observed word is combined with the leave-one-out mean of the other words in its spectrum, matched to the two closest ETM topic vectors, and aggregated into document-level evidence that adjusts the ETM posterior mean. Third, published 1.5-entmax replaces softmax to give exact document-topic zeros [12]. The ETM generator, multinomial likelihood, Gaussian variational family and analytic standard-normal Kullback–Leibler (KL) divergence remain intact. Channel balancing and entmax add no parameters;

contextual evidence adds one learned scalar. Controlled synthetic component comparisons test context and entmax separately, while the real-data canonical-versus-balanced ETM comparison isolates channel balancing.

This study makes three contributions:

1. a fully specified Contextual Sparse ETM that preserves explicit, inspectable Mass2Motif topic-word distributions;
2. a leakage-controlled benchmark covering data acquisition, spectral preprocessing, train-only token embeddings, document completion, MAG and SOS; and
3. truth-known synthetic validation together with three-seed real MSⁿLib validation of predictive fit, chemical utility, sparsity, global topic use and computational load.

2 Related work

Probabilistic and neural topic models. Latent Dirichlet allocation (LDA) models each document as a mixture of categorical topic-word distributions [4]. Neural Variational Document Models and Autoencoding Variational Inference for Topic Models (AVITM) established amortized variational inference, in which one encoder replaces separate iterative inference for every document. They also showed the practical importance of maintaining useful latent components [5, 6]. ETM subsequently parameterized topic-word scores through topic and word embedding inner products while retaining a probabilistic document model [7]. The present method uses this ETM generator and variational structure.

Short-document topic inference. Short-text models emphasize that individual documents provide limited co-occurrence evidence. The Biterm Topic Model aggregates word-pair statistics at corpus level [8], while the Neural Sinkhorn Topic Model uses embedding geometry and optimal transport, a geometry-aware distribution-matching objective, to improve neural topic modeling of short documents [9]. Empirical studies further show that document length changes the relationship between topic-model hyperparameters and evaluation measures [10]. Contextual Sparse ETM addresses the same statistical regime while preserving per-spectrum multi-topic mixtures and the ETM likelihood.

Sparse probability mappings. Sparsemax and the entmax family replace dense softmax probabilities with probability-vector mappings that can return exact zeros [11, 12]. 1.5-entmax lies between softmax and sparsemax and remains differentiable wherever its probabilities are non-zero. In the proposed model it acts only on document-topic logits; the topic-word decoder remains probabilistic and dense within each spectral channel.

Mass-spectrometry representation and evaluation. MS2LDA introduced the spectrum-as-document formulation [1]. MS2LDA 2.0 scaled the workflow through Tomotopy and introduced Mass2Motif Annotation Guidance (MAG) and the substructure overlap score (SOS) for chemical evaluation [2]. MSⁿLib provides the open reference-standard spectra used here [3], and Spec2Vec supplies the downstream retrieval representation used by MAG [13]. These established components define the domain task and its chemical evidence surface.

3 Materials and methods

3.1 MSⁿLib source, acquisition and loading

MSⁿLib is an open reference-standard library generated from seven compound collections using automated multi-stage fragmentation acquisition and processing. The complete resource contains more than 2.3 million MSⁿ spectra, including more than 357,000 MS² spectra for 30,008 unique structures across both ionization modes [3]. We used the positive-ion-mode benchmark input distributed with the MS2LDA 2.0 paper rather than resampling the complete resource [2].

The frozen assets are distributed in Zenodo record 20179680 [18]. The acquisition script downloads `Data.zip` and `model_positive_mode.zip`, verifies the declared byte size and MD5 checksum, checks ZIP-archive integrity and paths, extracts through a staging directory, verifies required files, and writes an acquisition manifest. The topic-model input is `Data/Benchmark_MSn_Lib/Corinna_Library_filtered_positive.mgf`. The positive-ion-mode Spec2Vec model, embedding matrix and SQLite spectral-library database are used only for downstream MAG evaluation. Known structures therefore support split construction and chemical scoring but are not labels for topic-model fitting.

The streaming Mascot Generic Format (MGF) reader treats each `BEGIN IONS/END IONS` block as one spectrum, parses peak lists rather than trusting `NUM_PEAKS`, and requires a Universal Spectrum Identifier, feature identifier, SMILES molecular-structure string, InChIKey structure identifier and precursor mass-to-charge ratio (m/z). The RDKit cheminformatics toolkit reparses each SMILES string and verifies agreement between the

derived connectivity InChIKey and the deposited identifier. Figure 1 summarizes the complete hand-off from public data to model matrices.



Figure 1: Reproducible MSⁿLib ingestion. The Mascot Generic Format (MGF) file supplies spectral documents; verified Spec2Vec similarity-retrieval assets branch from the extraction only for leakage-filtered chemical annotation.

3.2 Dataset construction and leakage control

The input MGF contains 41568 positive-ion-mode spectra. Within each spectrum, finite peak intensities are divided by the largest peak intensity. Peaks outside $m/z \in (0, 2000]$ or normalized intensity $[0.01, 1]$ are removed. If more than 1,000 peaks remain, the most intense are retained, and spectra with fewer than five retained peaks are discarded. This deterministic procedure retains 38888 spectra.

Every physical peak generates a fragment word and, when the precursor-minus-peak mass exceeds 0.01 Da, a neutral-loss word. Both masses are rounded to two decimal places, producing tokens such as **frag@70.04** and **loss@162.05**, consistent with MS2LDA 2.0 [2]. A peak with normalized intensity I contributes $\text{round}(100I)$ copies of each applicable word. The resulting document is a count-valued bag of spectral words. Fragment and neutral-loss tokens from the same peak are additionally retained as one physical peak group for document-completion splitting.

To prevent structurally related compounds from crossing partitions, molecules are keyed by the connectivity block of the InChIKey. Cyclic molecules are grouped by their non-chiral Bemis–Murcko core ring-and-linker scaffold [15], while each acyclic connectivity group remains intact. The 28572 resulting groups are sorted by descending size and assigned deterministically toward 70/10/20 train, validation and test targets using seed 42. This produces 27222 training, 3889 validation and 7777 test spectra from 38465 connectivity groups, with zero compound or split-group overlap.

The vocabulary is built from training spectra only, requires each token to occur in at least three training spectra (document frequency at least three), preserves first-seen training order, and contains 21233 fragment and neutral-loss tokens. Every split is stored in the same fixed column order as a SciPy compressed sparse-row (CSR) count matrix. All learned parameters, token embeddings, vocabulary entries and co-occurrence statistics use only the training partition. The present study uses training and validation evidence; the test partition was not loaded or scored for the proposed model.

For predictive evaluation, validation spectra are deterministically divided 50/50 by whole physical peak groups. A fragment and its corresponding neutral loss therefore cannot appear on opposite sides. The observed half is renormalized without consulting the maximum intensity of the withheld half. Out-of-vocabulary withheld words are recorded but excluded from the likelihood numerator and denominator. The completion result contains 3888 eligible spectra and 1,277,983 in-vocabulary withheld token copies; the withheld out-of-vocabulary fraction is 3.14%.

Finally, each vocabulary token has a fixed 48-dimensional skip-gram negative-sampling (SGNS) coordinate trained only on the training documents [14]. For five epochs, eight count-weighted positive pairs are sampled per document. Every positive pair uses five negative words drawn with probability proportional to corpus frequency raised to 0.75. Source embeddings are initialized uniformly on $[-0.5/48, 0.5/48]$, context embeddings at zero, and the PyTorch SparseAdam optimizer uses learning rate 0.01 with batches of 8,192 pairs. Source and context embeddings are averaged and normalized after the fifth epoch. Molecular identifiers and structures do not enter this representation. Holding these coordinates fixed separates the geometry supplied to every ETM variant from the learned topic vectors and posterior, so component comparisons cannot be attributed to variant-specific word-embedding updates.

3.3 Contextual Sparse ETM

3.3.1 Model lineage and design rationale

Contextual Sparse ETM begins with the published Embedded Topic Model of Dieng et al. [7]. ETM supplies the complete generative and variational backbone: topic and word vectors share an embedding space, their inner products define an explicit topic–word distribution, a neural encoder parameterizes a Gaussian document posterior, and a multinomial likelihood with analytic Gaussian KL divergence trains the model end to end. We retain that backbone rather than introducing a different topic-model family.

The adaptations in Table 1 address properties specific to short tandem mass spectra. Fixed train-only SGNS coordinates provide the spectral-word geometry consumed by ETM. Channel balancing prevents unequal fragment and neutral-loss vocabulary sizes from setting their total decoder mass. Contextual evidence lets the other peaks in one short spectrum disambiguate each token. Finally, published 1.5-entmax supplies exact per-spectrum sparsity. Only the context scalar is learned in addition to the ETM parameters.

Table 1: Lineage of Contextual Sparse ETM. “Inherited” means retained from the published Embedded Topic Model (ETM). The canonical-versus-balanced real comparison isolates channel balancing; the synthetic component study isolates contextual evidence and entmax.

Component	Provenance	Purpose
ETM generator, encoder, Gaussian posterior, likelihood and KL divergence	Inherited [7]	Preserve a published probabilistic topic model with explicit β and amortized document inference.
Train-only skip-gram (SGNS) word coordinates	ETM input configuration [14]	Supply leakage-free fragment/loss co-occurrence geometry.
Fragment/loss-balanced β	Added; no parameters	Give the paired spectral channels equal aggregate probability mass.
Contextual top-2 posterior evidence	Added; one scalar	Use neighbouring peaks to disambiguate tokens in a short spectrum.
1.5-entmax document mixture	Published mapping [12]	Replace dense posterior softmax with exact document-topic zeros.

3.3.2 ETM generator and channel-balanced decoder

Let $d \in \{1, \dots, D\}$ index spectra, $w \in \{1, \dots, V\}$ index words and $k \in \{1, \dots, K\}$ index topics. Let $c_{dw} \geq 0$ be the raw spectral-word pseudo-count and

$$x_{dw} = \frac{c_{dw}}{\sum_v c_{dv}} \quad (1)$$

the normalized encoder input. Each word has a fixed SGNS vector $\rho_w \in \mathbb{R}^{48}$, and each topic has a learned ETM vector $\alpha_k \in \mathbb{R}^{48}$. The stored SGNS rows are unit-normalized before model construction. Throughout,

$$\text{normalize}(v) = \frac{v}{\max(\|v\|_2, \epsilon)}, \quad \text{center}(v) = v - \frac{1}{K} \mathbf{1} \mathbf{1}^\top v, \quad \epsilon = 10^{-12}.$$

The topic-word distribution follows the ETM inner-product construction [7]. Fragment and neutral-loss channels are normalized separately so vocabulary size cannot determine their aggregate mass. With V_F and V_L the two channel vocabularies,

$$\beta_{kw} = \begin{cases} \frac{1}{2} \frac{\exp(\alpha_k^\top \rho_w)}{\sum_{v \in V_F} \exp(\alpha_k^\top \rho_v)}, & w \in V_F, \\ \frac{1}{2} \frac{\exp(\alpha_k^\top \rho_w)}{\sum_{v \in V_L} \exp(\alpha_k^\top \rho_v)}, & w \in V_L. \end{cases} \quad (2)$$

Thus $\sum_w \beta_{kw} = 1$, each topic assigns exactly one half of its probability to fragments and one half to neutral losses, and token rankings within each channel remain learned. This channel constraint adds no parameters.

The ordinary ETM encoder is a two-layer rectified network,

$$e_d = \text{ReLU}(W_2 \text{ReLU}(W_1 x_d + b_1) + b_2), \quad (3)$$

$$\mu_d = W_\mu e_d + b_\mu, \quad \log \sigma_d^2 = W_\sigma e_d + b_\sigma, \quad (4)$$

with two hidden layers of width 800.

3.3.3 Contextual evidence, sparse posterior and training

Short spectra provide most of their disambiguating information through the other words observed alongside a token. Define normalized word and topic vectors $\hat{\rho}_w = \text{normalize}(\rho_w)$ and $\hat{\alpha}_k = \text{normalize}(\alpha_k)$. For every non-zero word in spectrum d , its weighted leave-one-out context is

$$\bar{\rho}_{d \setminus w} = \frac{\sum_v x_{dv} \hat{\rho}_v - x_{dw} \hat{\rho}_w}{\max(1 - x_{dw}, \epsilon)}. \quad (5)$$

The contextual word representation is

$$h_{dw} = \text{normalize}(\hat{\rho}_w + c \bar{\rho}_{d \setminus w}), \quad (6)$$

where c is one learned scalar.

The word is scored against the same topic vectors that define the decoder:

$$a_{dwk} = h_{dw}^\top \hat{\alpha}_k / \tau_r, \quad \tau_r = 1. \quad (7)$$

Let $\mathcal{T}_{dw}$ contain the two largest scores. The local contextual assignment is

$$q_{dwk} = \begin{cases} \frac{\exp(a_{dwk})}{\sum_{j \in \mathcal{T}_{dw}} \exp(a_{dwj})}, & k \in \mathcal{T}_{dw}, \\ 0, & \text{otherwise.} \end{cases} \quad (8)$$

Count-weighted aggregation yields one document evidence distribution,

$$r_{dk} = \sum_w x_{dw} q_{dwk}. \quad (9)$$

Because every local assignment and each x_d are normalized, r_d lies on the topic probability simplex: its entries are non-negative and sum to one. The implementation renormalizes its row sum once more to remove finite-precision accumulation error.

The document evidence is converted to a bounded, row-centred posterior offset:

$$o_d = \text{center}[\log(r_d + \frac{1}{K})], \quad \tilde{\mu}_d = \mu_d + o_d. \quad (10)$$

The fixed $1/K$ pseudocount bounds evidence for topics outside the local top-2 assignments, while centring makes uniform evidence an exact no-op. The variational sample and document mixture are

$$z_d = \tilde{\mu}_d + \exp(\frac{1}{2} \log \sigma_d^2) \odot \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I), \quad (11)$$

$$\theta_d = \text{entmax}_{1.5}(z_d). \quad (12)$$

1.5-entmax supplies exact zeros while preserving a normalized mixture. More explicitly, it computes

$$\theta_{dk} = \left[\frac{z_{dk}}{2} - \tau_d \right]_+^2, \quad \sum_k \left[\frac{z_{dk}}{2} - \tau_d \right]_+^2 = 1, \quad (13)$$

where the threshold τ_d is determined from the sorted logits. The code then divides by the finite-precision row sum, which preserves all exact zeros. At deterministic inference, $z_d = \tilde{\mu}_d$.

The conditional word probability remains the usual topic mixture,

$$p(w | d) = \sum_{k=1}^K \theta_{dk} \beta_{kw}. \quad (14)$$

The stochastic loss supplied to Adam is the raw pseudo-count multinomial reconstruction (up to its parameter-independent combinatorial constant) plus the analytic standard-normal Gaussian KL divergence. One reparameterized Gaussian draw is used per document and minibatch update:

$$\mathcal{L}_{\text{rec}} = -\frac{1}{D} \sum_d \sum_w c_{dw} \log p(w | d), \quad (15)$$

$$\mathcal{L}_{\text{KL}} = -\frac{1}{2D} \sum_d \sum_k (1 + \log \sigma_{dk}^2 - \tilde{\mu}_{dk}^2 - \exp(\log \sigma_{dk}^2)), \quad (16)$$

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \mathcal{L}_{\text{KL}}. \quad (17)$$

Adam additionally applies coupled weight decay $\lambda = 1.2 \times 10^{-6}$ to all learned parameters, i.e. it adds $\lambda \phi$ to the gradient for parameter vector ϕ . No auxiliary co-occurrence, topic-separation or assignment loss is used.

Figure 2 shows how contextual evidence adjusts the posterior while the ETM generator and explicit topic-word output remain intact.

With vocabulary size V , hidden width H , topics K and embedding dimension L , the learned-parameter count is

$$VH + H + H^2 + H + 2(KH + K) + KL + 1.$$

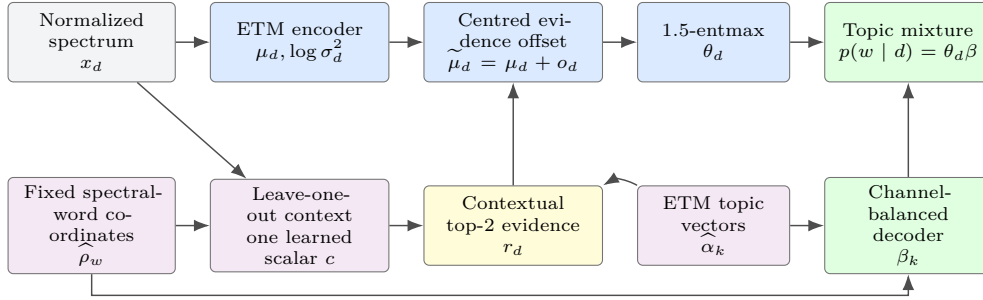


Figure 2: Contextual Sparse Embedded Topic Model (ETM). Contextual token assignments use the same fixed word and learned topic coordinates as the ETM topic-word decoder, then adjust the Gaussian document posterior before 1.5-entmax produces a sparse topic-probability vector. Solid arrows denote computations used at both training and inference.

For $V = 21,233$, $H = 800$, $K = 1000$ and $L = 48$, the model has 19,278,001 learned parameters, compared with 19,278,000 for the channel-balanced ETM; fixed ρ rows are not counted. The only addition is c , which converged to 0.1772 in the representative run. Models are trained for 120 fixed epochs with Adam, learning rate 0.005 , weight decay 1.2×10^{-6} and batch size 256. PyTorch deterministic algorithms are enabled, non-finite objectives or gradients terminate execution, and six central processing unit (CPU) threads support training on an NVIDIA CUDA-compatible graphics processing unit (GPU).

At validation, the observed half-spectrum is encoded deterministically for document completion, while the full spectrum is encoded for topic-inventory and chemical analyses. All 1000 topic-word distributions are exported for MAG. Training and inference use the same equations; no spectrum-specific optimization loop is required.

3.4 Experimental design and comparators

The simulator contains 18 planted motifs with paired fragment and neutral-loss words. Each spectrum contains one to three generating motifs. Training-only vocabulary construction and SGNS match the real workflow, and intensities use the same raw pseudo-count reconstruction. The primary experiment uses 800 training and 160 validation spectra, fits $K = 36$ topics, and averages seeds 11, 23 and 37. A high-overcompleteness experiment, in which fitted topics greatly outnumber planted motifs, fixes seed 11 and increases the fitted inventory to $K = 128$ while retaining 18 planted motifs.

The synthetic endpoints are completion NLL, cosine recovery of the planted topic-word matrix, cosine recovery of the planted document mixtures, dominant motif accuracy, median effective topics, materially active topics and unique top-1 topics. Dominant motif accuracy compares the largest fitted mixture component with the largest planted component; a top-1 topic is the fitted topic with the largest weight in one spectrum. Optimal one-to-one matching aligns fitted topics to planted motifs before recovery is calculated.

For real-data validation, four methods are fitted to the same training and validation matrices:

1. canonical ETM: fixed SGNS coordinates, global topic-word softmax and dense document-topic softmax;
2. fragment/loss-balanced ETM: the same model with separate 50/50 fragment and neutral-loss decoder normalization;
3. Contextual Sparse ETM: the balanced decoder plus contextual top-2 posterior evidence and 1.5-entmax document mixtures; and
4. Tomotopy 0.13.0 LDA: the established MS2LDA-style comparator [17].

The first three rows therefore follow the model lineage from published ETM to the channel-balanced decoder and then the complete contextual sparse posterior. The two ETM baselines use the same 48-dimensional embeddings, two-layer 800-unit encoder, $K = 1000$, raw pseudo-count objective, Adam settings, batch size, 120 epochs and training seed 7043. Tomotopy uses $K = 1000$, $\alpha = 0.6$, $\eta = 0.1$, six workers, parallel scheme 3 and 750 training iterations; validation inference uses 100 iterations and parallel scheme 1.

Initialization robustness is assessed by repeating the proposed model with training seeds 7043, 23 and 37. The data split, vocabulary, SGNS embeddings, configuration, epoch count and chemical evaluation remain fixed. These repeats measure sensitivity to model initialization and minibatch order, not uncertainty over data splits.

3.5 Evaluation measures

Document-completion negative log-likelihood scores withheld in-vocabulary token copies after observing the other half of each spectrum:

$$\text{NLL}_{\text{comp}} = -\frac{\sum_{d,w} c_{dw}^{\text{held}} \log \left(\sum_k \hat{\theta}_{dk}^{\text{obs}} \beta_{kw} \right)}{\sum_{d,w} c_{dw}^{\text{held}}}. \quad (18)$$

Lower values indicate better prediction of unseen spectral evidence.

Per-spectrum sparsity is measured by exact support size and the entropy-effective topic count

$$K_{\text{eff}}(\theta_d) = \exp \left[-\sum_k \theta_{dk} \log \theta_{dk} \right]. \quad (19)$$

Writing $\bar{\theta}_k = D^{-1} \sum_d \theta_{dk}$, the corpus-effective inventory is $\exp[-\sum_k \bar{\theta}_k \log \bar{\theta}_k]$. Global inventory use is also summarized by the number of topics that are the largest mixture component for at least one spectrum (unique top-1 topics), topics with mean mixture weight above 0.0005, maximum mean topic weight and cosine similarity between topic-word probability vectors. A duplicate component is catastrophic only if at least half of the 1,000 topics join one connected component in the cosine-similarity graph at threshold 0.999.

Chemical utility is evaluated with the MS2LDA 2.0 Mass2Motif Annotation Guidance (MAG) and sub-structure overlap score (SOS) framework [2]. Every Spec2Vec-library row sharing a validation connectivity InChIKey is removed before constructing the FAISS vector-similarity search index. The evaluated molecule therefore cannot be retrieved as its own annotation evidence.

For each topic, the 20 highest-probability fragment or neutral-loss words form a Mass2Motif pseudo-spectrum: a spectrum-like list synthesized from the learned topic probabilities. MAG searches up to 1,000 neighbours, retains at most ten unique molecules, clusters and refines them at spectral cosine-similarity threshold 0.9, and optimizes the motif against the selected cluster. A topic is *optimized* when at least one fragment or loss survives this procedure.

Clustered MAG molecules are converted to binary MACCS structural-key fingerprints, whose bits indicate predefined molecular features [16]. A bit is retained in consensus fingerprint A_k when it appears in at least 80% of the cluster. A validation molecule i is associated with topic k when its full-spectrum topic weight satisfies $\theta_{ik} \geq 0.5$. Repeated spectra with the same connectivity InChIKey are collapsed. With B_i the associated molecule fingerprint,

$$s_{ki} = \frac{|A_k \cap B_i|}{|A_k|}, \quad \text{SOS}_k = \frac{1}{|C_k|} \sum_{i \in C_k} s_{ki}. \quad (20)$$

The implementation assigns $s_{ki} = 0$ if A_k contains no retained bits. An optimized motif is *SOS-evaluable* when it has at least one associated unique molecule with a valid fingerprint. A *useful* motif is evaluable with SOS ≥ 0.6 , combining the intermediate and high bands defined by MS2LDA 2.0. SOS measures containment of a consensus fingerprint; it is not an atom mapping, fragmentation-pathway proof or expert-confirmed annotation.

4 Results

4.1 Synthetic component comparisons isolate complementary context and sparsity

The $K = 36$ experiment is a controlled comparison of the two posterior changes. Starting from the channel-balanced ETM with softmax, adding contextual top-2 evidence while retaining softmax improves both planted topic-word (β) and document-mixture (θ) recovery, improves completion NLL, and increases the number of planted topics used as document winners. Replacing softmax with 1.5-entmax without contextual evidence does create sparse mixtures, but it reduces both recovery measures and uses fewer planted components. Thus sparsity alone is not the successful mechanism.

Combining contextual evidence with 1.5-entmax yields the strongest mean β and θ recovery, retains a median of two effective topics per spectrum, and uses 13 distinct topics as winners, consistent with the one-to-three motif simulator. Its completion NLL is slightly worse than the context-plus-softmax variant, so the evidence supports complementarity rather than uniform dominance: context preserves and disambiguates the topic inventory, while entmax converts that evidence into exact sparse support.

Table 2: $K = 36$ synthetic results averaged over seeds 11, 23 and 37. NLL is held-out completion negative log-likelihood (lower is better). Beta and theta are cosine recovery of the planted topic-word and document-topic matrices; median eff. is the median entropy-effective topic count. Active topics have mean corpus weight above 0.005, and unique top-1 counts distinct document winners.

Formulation	NLL	Beta	Theta	Median eff.	Active	Unique top-1
Balanced ETM + softmax	6.4710	0.3052	0.4868	3.59	8.00	6.67
Balanced ETM + 1.5-entmax	6.5977	0.2566	0.3973	1.88	5.00	4.67
Contextual top-2 evidence + softmax	6.2438	0.4423	0.7066	3.22	13.33	11.67
Contextual Sparse ETM	6.2892	0.4648	0.7513	2.00	14.00	13.00

The high-overcompleteness experiment strengthens this interpretation. At $K = 128$, entmax without context has exact support two but uses only three of the 18 planted motifs as winners. Contextual Sparse ETM keeps exact support two, recovers all 18 planted motifs as distinct winners, achieves beta/theta recovery 0.6638/0.9556 and dominant-motif accuracy 0.9313, and gives the lowest completion NLL of the three high- K formulations.

Table 3: Seed-11 synthetic stress with 18 planted motifs and 128 fitted topics. NLL is completion negative log-likelihood; beta and theta are planted-matrix recovery; accuracy compares the dominant fitted and planted motif. Eff. is the entropy-effective topic count, support counts exactly non-zero document-topic weights, and unique counts distinct document winners.

Formulation	NLL	Beta	Theta	Accuracy	Eff.	Support	Unique
Balanced ETM + softmax	6.4175	0.3696	0.5782	0.5438	3.67	128	7
Balanced ETM + 1.5-entmax	6.7375	0.2344	0.2965	0.2625	1.72	2	3
Contextual Sparse ETM	6.1276	0.6638	0.9556	0.9313	1.45	2	18

4.2 Real-data motif discovery and initialization stability

On the real MSⁿLib validation split, Contextual Sparse ETM produces 445 SOS-evaluable and 289 useful motifs. The corresponding counts are 130/79 for canonical ETM, 166/104 for fragment/loss-balanced ETM and 206/138 for Tomotopy. Figure 3 shows that the gain occurs after chemical evaluation rather than merely through creating a large unfiltered topic inventory.

The canonical-versus-balanced ETM rows provide the real-data controlled comparison of the decoder change. Channel balancing raises optimized motifs from 609 to 911, evaluable motifs from 130 to 166, and useful motifs from 79 to 104, while mean SOS changes from 0.6388 to 0.6298 and completion NLL from 8.6907 to 8.7661. It therefore broadens the candidate inventory modestly but does not explain the complete model’s 445 evaluable and 289 useful motifs. The larger gain appears only after adding the contextual sparse posterior tested in the synthetic component comparisons.

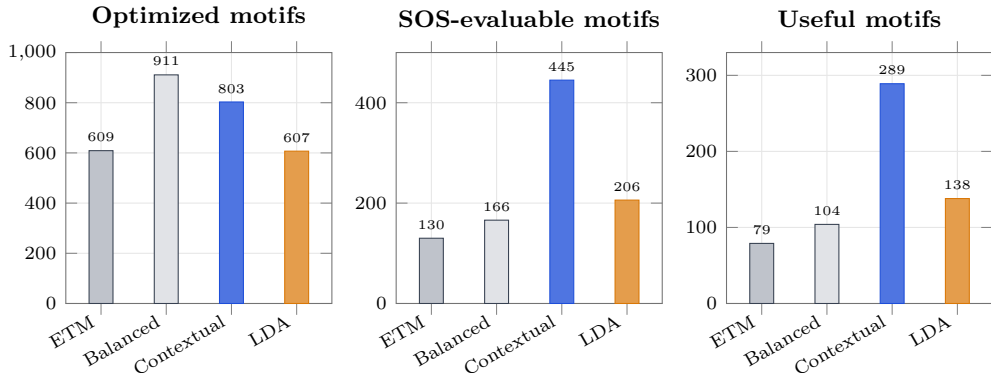


Figure 3: Validation motif inventory after successive chemical filters. ETM is the canonical Embedded Topic Model with fixed train-only spectral-word coordinates and dense softmax document mixtures. Balanced is the same ETM with separate 50/50 fragment and neutral-loss decoder normalization. Contextual is the complete Contextual Sparse ETM, which adds contextual top-2 posterior evidence and 1.5-entmax to Balanced. LDA is Tomotopy 0.13.0 latent Dirichlet allocation. Optimized means that Mass2Motif Annotation Guidance (MAG) retains at least one fragment or loss; SOS-evaluable means that the substructure overlap score can be computed from at least one associated molecule; useful means SOS at least 0.6. All four models use $K = 1000$ and the same frozen training/validation split and chemical-evaluation protocol.

Table 4: Validation comparison. SOS is the substructure overlap score and NLL is held-out completion negative log-likelihood, for which lower is better. Bold values are the numerically strongest entry in each metric; no single metric defines overall model quality.

Model	Optimized	Evaluable	Useful	Mean SOS	Median SOS	NLL ↓
Canonical ETM	609	130	79	0.6388	0.6313	8.6907
Fragment/loss-balanced ETM	911	166	104	0.6298	0.6329	8.7661
Contextual Sparse ETM	803	445	289	0.6472	0.6579	9.5429
Tomotopy LDA	607	206	138	0.6761	0.6854	9.6622

Figure 4 places conditional chemical quality against the size of the evaluable inventory. It makes the central comparison visible: Contextual Sparse ETM moves far to the right of all three comparators while retaining mean SOS above both ETM baselines; Tomotopy occupies the higher-quality but lower-breadth corner.

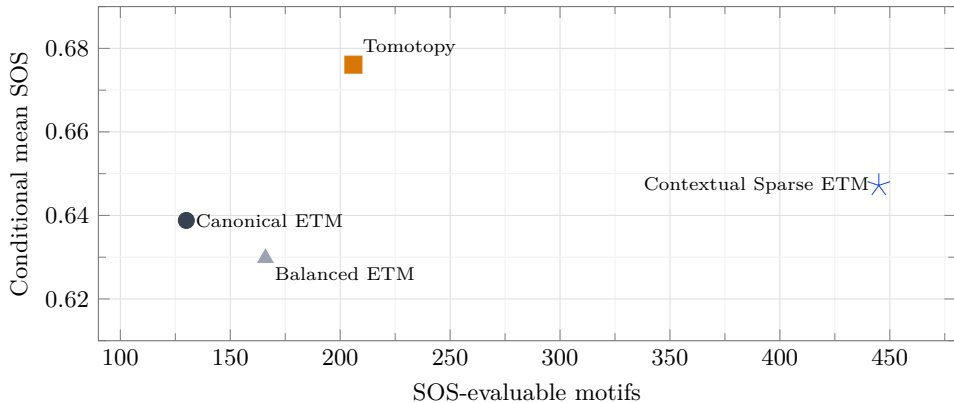


Figure 4: Discovery breadth versus conditional chemical quality on the frozen validation split. Horizontal position counts motifs for which the substructure overlap score (SOS) can be evaluated; vertical position averages SOS only over that model-specific set. The axes therefore answer complementary questions rather than forming a single combined score. Points compare canonical and fragment/loss-balanced Embedded Topic Models (ETMs), the complete Contextual Sparse ETM and Tomotopy LDA.

The proposed model converts 55.4% of optimized motifs into evaluable motifs and 36.0% into useful motifs. Its 803 optimized motifs are fewer than the balanced ETM’s 911, but far more survive spectrum association

and SOS evaluation. Relative to Tomotopy, the proposed model adds 239 evaluable and 151 useful motifs and also lowers completion NLL from 9.6622 to 9.5429.

The result is not uniform dominance. Tomotopy’s smaller evaluable set has higher mean and median SOS (0.6761/0.6854) than the proposed model (0.6472/0.6579). Canonical and balanced ETM have lower completion NLL but produce substantially fewer evaluable and useful motifs. The primary result is therefore increased discovery breadth with an explicit conditional-quality and predictive-fit trade-off.

Changing only initialization and minibatch order tests whether this discovery gain depends on one fitted checkpoint. The three runs give 787–803 optimized, 439–453 evaluable and 274–289 useful motifs. Mean SOS spans 0.6376–0.6473 and completion NLL spans only 9.5394–9.5460. Median exact support is six for every run, and 813–828 topics become the largest component for at least one validation spectrum. No run contains a catastrophic duplicate component.

Table 5: Same-split training-seed stability. SOS is the substructure overlap score, NLL is completion negative log-likelihood and median eff. is the median entropy-effective topic count; unique counts distinct document winners. The final row reports the descriptive sample mean and standard deviation for three initializations, not population uncertainty.

Seed	Optimized	Evaluable	Useful	Mean SOS	Median SOS	NLL	Median eff.	Unique
7043	803	445	289	0.6472	0.6579	9.5429	3.699	828
23	791	453	274	0.6376	0.6395	9.5460	3.702	816
37	787	439	275	0.6473	0.6477	9.5394	3.714	813
Mean $\pm$ SD	793.7 $\pm$ 8.3	445.7 $\pm$ 7.0	279.3 $\pm$ 8.4	0.6440 $\pm$ 0.0056	0.6484 $\pm$ 0.0092	9.5428 $\pm$ 0.0033	3.705 $\pm$ 0.008	819.0 $\pm$ 7.9

The tight NLL and sparsity ranges show that the mechanism is not specific to one initialization. Chemical scores vary more than predictive fit: seed 23 produces the largest evaluable inventory and the lowest mean SOS. This pattern is consistent with a breadth–quality trade-off and motivates reporting motif counts together with the SOS distribution rather than reducing performance to one average.

4.3 Sparse spectrum mixtures, broad topic use and computational cost

The representative run has median 3.70 entropy-effective topics and median exact support 6 per validation spectrum; 95% of spectra have support at most 13. At corpus level, 828 of 1,000 topics win at least one spectrum, 535 exceed mean usage 0.0005 and the corpus-effective topic count is 538.4. The largest connected component at topic–word cosine similarity ≥ 0.999 contains only 2 topics.

Table 6: Validation sparsity and inventory diagnostics. Eff. is the entropy-effective topic count; support counts exactly non-zero mixture weights; unique top-1 counts distinct document winners; and active topics have mean weight above 0.0005. Corpus eff. applies the entropy-effective count to mean corpus weights, and nearest beta is mean nearest-neighbour cosine similarity between topic–word probability vectors.

Model	Median eff.	Median support	Unique top-1	Active	Corpus eff.	Nearest beta
Canonical ETM	43.79	–	260	269	344.6	0.321
Fragment/loss-balanced ETM	46.72	–	260	275	355.7	0.400
Contextual Sparse ETM	3.70	6	828	535	538.4	0.153

The ETM baselines spread each spectrum across roughly 44–47 effective topics while using only 260 unique winners. Contextual Sparse ETM reverses both properties: individual spectra are sparse, yet more than 800 topics receive a top-1 assignment somewhere in validation. Its maximum mean topic usage is only 0.0143, and mean nearest-topic topic–word cosine similarity is 0.1529, providing additional evidence that sparsity does not come from a single dominant or globally duplicated topic group.

Computationally, the 120-epoch optimization loop takes 13.7 ± 0.8 minutes. With the trained model already resident on the GPU, deterministic inference processes 23,426 full validation spectra per second. Peak memory actively allocated by PyTorch and memory reserved by its CUDA cache are 0.821 and 1.053 GB, respectively. Peak process memory, including CPU memory, is 2.879 GB, while at least 12.43 GB of system memory remains available.

These measurements show that the method fits comfortably on the evaluated GPU system. They are implementation and hardware measurements rather than algorithm-independent complexity claims. The ETM and Tomotopy reference fits were executed on CPU, so their recorded wall times are not used for a direct training-speed ranking.

5 Discussion

5.1 Mechanism and model economy

The proposed model separates two roles. The ETM generator gives every topic an explicit fragment/loss distribution, preserving the object required for Mass2Motif inspection and MAG. The posterior adaptation asks a different question: which topics are supported by the words observed together in this particular short spectrum? Leave-one-out context lets a common fragment or loss contribute differently when its neighbouring evidence changes. Restricting each word to its two strongest topic matches prevents weak similarity to hundreds of topics from accumulating into a diffuse document signal.

The centred log offset does not replace the variational encoder. It supplies a bounded evidence correction in the encoder’s own topic coordinates. Uniform contextual evidence leaves the posterior unchanged, and topics outside local top-2 assignments retain finite evidence through the fixed $1/K$ pseudocount. Finally, 1.5-entmax converts the corrected latent representation into exact support. The synthetic results show why these operations are complementary: contextual evidence improves recovery and topic use, while sparse projection controls the document support.

The complete model can be explained through four objects: fixed spectral-word coordinates, learned ETM topic coordinates, one contextual evidence distribution per spectrum and explicit θ and β probability vectors. Channel balancing has a direct domain interpretation and no learned parameters. The posterior adaptation adds only the context scalar c ; assignment temperature, evidence strength, top-2 support and pseudocount are fixed. Consequently, the method remains recognizable as an ETM rather than replacing the generator with a separate clustering system.

Interpretability nevertheless remains statistical. A high-probability fragment/loss pattern is a Mass2Motif hypothesis, and SOS measures containment of a MAG-derived consensus fingerprint. Chemical naming, atom-level mapping and fragmentation-mechanism claims still require expert review and, ideally, orthogonal experimental evidence.

5.2 Breadth–quality trade-offs and validation priorities

Optimized, evaluable and useful motif counts measure how much of the learned inventory becomes available for chemical interpretation. Mean and median SOS measure chemical containment conditional on passing the evaluability filter. A model can therefore expose more useful motifs in absolute terms while having a lower average over that larger set. Contextual Sparse ETM exhibits exactly this pattern relative to Tomotopy: it more than doubles evaluable and useful counts, whereas Tomotopy retains higher conditional SOS. The breadth–quality plot makes this conditional denominator explicit and avoids treating an average over 445 motifs as though it summarized the same set as an average over 206.

Completion NLL provides a third axis. The proposed model predicts withheld evidence better than Tomotopy but not as well as the dense ETM baselines. This trade-off is plausible because sparse, high-confidence document mixtures impose a stronger allocation constraint than dense mixtures. The results do not support a blanket superiority claim; they support the narrower conclusion that the posterior adaptation materially increases chemically assessable motif breadth while preserving sparse assignments and a broad global inventory.

These axes correspond to different uses. Completion NLL tests probabilistic generalization, support diagnostics test whether spectrum explanations remain compact, and the MAG and SOS measures determine how much chemically interpretable material reaches a downstream analyst. For discovery, the larger useful set is valuable even when it should subsequently be ranked or filtered by conditional chemical evidence. Reporting all three prevents a favourable result on one axis from concealing collapse on another.

The next decisive experiment is generalization beyond the present frozen split. A second scaffold split or an external spectral library would test sensitivity to data composition. Once the method and interpretation are independently frozen, the untouched test partition can provide a single confirmatory evaluation. Negative-mode spectra, biological samples and instrument-shifted data would establish a broader domain of validity. These extensions should retain the same reporting contract: predictive fit, per-spectrum sparsity, global inventory use, MAG-retrieved and SOS-evaluable motif breadth, and conditional chemical quality.

6 Limitations

This is a validation study on one positive-ion-mode MSⁿLib benchmark. The three training seeds change initialization and minibatch order but not the data split, vocabulary, topic count or hyperparameters. Their sample standard deviations are descriptive and should not be interpreted as confidence intervals over future datasets. The test split remains untouched for the proposed model.

The synthetic simulator supplies known truth but cannot represent the complete chemical, instrumental and noise structure of measured spectra. Its recovery metrics establish mechanism behaviour under controlled conditions rather than biological validity.

The observation model treats rounded relative intensities as multinomial pseudo-counts. Those copies are weights, not independent ion observations, and their absolute scaling changes the balance between reconstruction and KL divergence. The fixed 50/50 fragment/loss mass is an explicit domain prior that removes vocabulary-size bias but may not be optimal for every acquisition regime.

MAG and SOS depend on the Spec2Vec reference library, leakage filter, retrieval and clustering settings, MACCS fingerprint bits, 80% consensus threshold and $\theta \geq 0.5$ molecule-association threshold. SOS is an asymmetric containment measure and does not establish an exact substructure. Reference library coverage can also influence which topics become evaluable.

Finally, the runtime measurements describe one implementation and GPU system. The dense ETM encoder scales with vocabulary size, and peak memory or throughput will change with hardware, batch size, numerical precision and software versions.

7 Data, code and reproducibility

The benchmark input and positive-ion-mode annotation assets are available from Zenodo record 20179680 [18]. The repository records the exact data preparation protocol, model implementation, compact result packages, per-topic chemical scores, epoch-level training records, environment manifest and SHA-256 file checksums. Test artifacts for the proposed model are explicitly excluded from the current evidence boundary.

Every numerical fragment in this report is generated from frozen machine-readable evidence files in JavaScript Object Notation (JSON) and comma-separated values (CSV) formats:

```
python -m scripts.generate_routing_etm_report
python -m scripts.verify_routing_etm_checkpoint
python -m scripts.verify_routing_etm_stability
```

The first command validates cross-file identities and regenerates the LaTeX tables. The two verification commands check the primary and three-seed checkpoint packages without requiring retained large local arrays. Full artifact and input hashes can additionally be checked on the originating host. The focused equation-code test in `benchmarks/neural_ms2lda/tests/test_routing_etm.py` independently reconstructs the channel-balanced β , contextual top-2 evidence, centred posterior mean, analytic KL divergence, threshold form of 1.5-entmax, raw-count reconstruction and learned-parameter count, then compares each quantity with the implementation.

8 Conclusion

Contextual Sparse ETM is a direct extension of the published ETM rather than a separate topic-model family. It retains ETM’s embedded topic-word generator, Gaussian variational encoder, multinomial objective and analytic KL divergence. Equal fragment/loss decoder mass supplies the domain constraint; shared word/topic geometry supplies contextual top-2 evidence; and 1.5-entmax produces exact document-topic support. The complete adaptation adds only one learned scalar.

On scaffold- and compound-disjoint MSⁿLib validation, the model produces substantially more motifs eligible for SOS and more useful motifs than canonical ETM, fragment- and loss-balanced ETM and Tomotopy, while retaining approximately 3.7 effective topics per spectrum and more than 800 unique validation winners. The result is stable across three initializations. Tomotopy retains higher conditional SOS, and dense ETM retains better completion likelihood, so the scientific conclusion is a reproducible discovery-breadth advantage with quantified quality and predictive trade-offs. External data and the untouched test partition are required for confirmatory generalization.

A Exact hyperparameters

Table 7: Frozen settings for the reported model and evaluation.

Quantity	Value
Data split seed	42
Training seeds	7043, 23, 37
Topics	1000
Vocabulary	21,233 train-only fragment/loss tokens
Token coordinates	48-dimensional train-only skip-gram negative-sampling (SGNS); fixed
Encoder	21,233 $\rightarrow$ 800 $\rightarrow$ 800; rectified linear unit (ReLU)
Variational outputs	1,000-dimensional μ and $\log \sigma^2$
Topic-word decoder	Embedded Topic Model (ETM) inner products; 50/50 fragment/loss mass
Contextual evidence	leave-one-out context; top-2; temperature 1.0
Evidence pseudocount	fixed $1/K$
Numerical normalization floor	10^{-12}
Additional learned parameters	one context scalar
Document-topic probability transform	1.5-entmax
Reconstruction	raw intensity pseudo-count multinomial
Prior and KL divergence	standard-normal analytic Gaussian Kullback–Leibler divergence
Optimizer	Adam
Learning rate; weight decay	0.005; 1.2×10^{-6}
Batch size; epochs	256; 120
Device; CPU threads	NVIDIA CUDA GPU; 6
Mass2Motif Annotation Guidance (MAG) query	top 20 words; search 1000; 10 unique molecules
MAG and substructure overlap score (SOS) thresholds	membership ≥ 0.5 ; cluster cosine 0.9; consensus 0.8

B Equation–implementation correspondence

Table 8: Map from the mathematical description to executable repository code.

Report component	Implementation
Balanced ETM and β	<code>sparse_etm.py: BalancedSparseETM</code>
Contextual token evidence	<code>routing_etm.py: routing_evidence</code>
Posterior offset and KL divergence	<code>routing_etm.py: encode</code>
1.5-entmax θ	<code>sparse_etm.py: transform_theta</code>
Reconstruction objective	<code>sparse_etm.py: sparse_reconstruction_loss</code>
Real training and inference	<code>scripts/run_routing_etm_real.py</code>
Synthetic design	<code>scripts/run_routing_etm_campaign.py</code>
Document completion	<code>objectives.py: completion_metrics</code>
MAG and SOS	<code>mag.py; chemical.py</code>
Equation–code correspondence test	<code>tests/test_routing_etm.py</code>
Checkpoint verification	<code>scripts/verify_routing_etm_checkpoint.py</code>
Stability verification	<code>scripts/verify_routing_etm_stability.py</code>

C Metric interpretation

Metric	Definition	Reading
Completion negative log-likelihood (NLL)	Count-weighted log loss on withheld in-vocabulary words after observing the other physical peak groups.	Lower is better predictive fit.
Optimized motifs	Topics for which Mass2Motif Annotation Guidance (MAG) retains at least one fragment or neutral loss.	Breadth of annotatable topic spectra.
Substructure-overlap-score (SOS)-evaluable motifs	Optimized topics with at least one unique associated validation molecule and a valid fingerprint.	Inventory eligible for chemical scoring.
Mean/median SOS	Containment of the MAG consensus fingerprint in associated molecules, averaged first within motif and then across motifs.	Higher is better conditional consistency.
Useful motifs	SOS-evaluable motifs with SOS at least 0.6.	Absolute count in intermediate or high bands.
Effective topics	Exponentiated entropy of one spectrum’s topic mixture.	Lower means a more concentrated mixture.
Unique top-1 topics	Topics that are the largest mixture component for at least one validation spectrum.	Breadth of document-level topic use.

Table 9: Interpretation of the main predictive, chemical and inventory measures. Counts and conditional averages answer different questions.

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